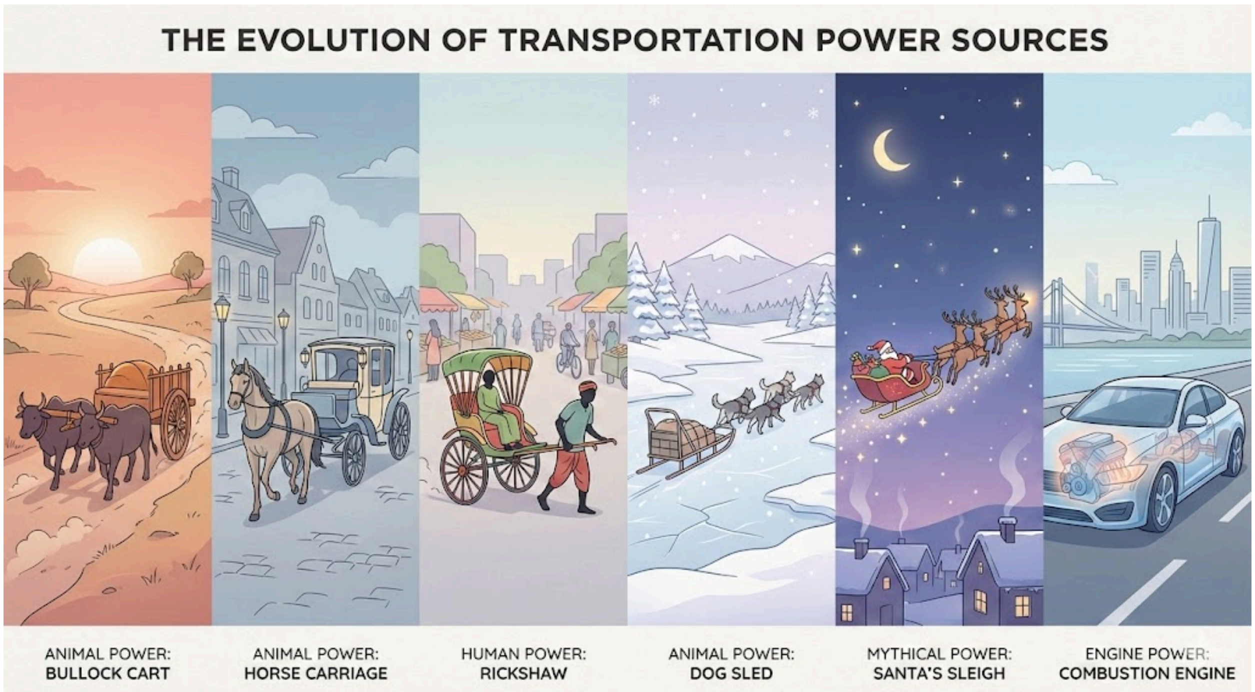
Engine

So let's start from the basics. Every automotive vehicle needs something to make it move. Something that pushes it or pulls it through its surroundings. In the olden days, this job was done by bulls, horses, dogs, and in some places, even people. But now we use engines to do that work. Some people even say the engine is the heart of the vehicle, because without it, nothing really happens.



There are many types of engines. You have internal combustion engines, which themselves have many types. You also have jet turbine engines, rotary engines, and even an electric motor in a small toy car can be considered an engine. The basic idea is the same. An engine converts some form of energy into motion.

As you can guess, engines can't function just on their own. They need energy from some source. Just like how a household appliance needs electricity to run, an engine also needs energy. For most engines, that energy comes from fuel. And to release energy from fuel, we usually need oxygen. When fuel and oxygen come together and burn, energy is released. That energy is what we use to create a pushing force.

Now let's focus on the most common engine you see around you every day, the car engine, also called an internal combustion engine, or IC engine for short. "Internal combustion" simply means the fuel burns inside the engine, not outside. There are two major types of IC engines. One is the spark ignition engine, also called an SI engine. The other is the compression ignition engine, also called a CI engine. Petrol engines are usually SI engines, and diesel engines are usually CI engines.

An IC engine has many parts, but let’s only look at the main ones for now. There is the cylinder head, the engine block, the piston, the connecting rod, and the crankshaft. Don’t worry about remembering everything at once. We’ll go step by step.

The cylinder head sits on top of the engine. Inside it, there are passages or holes called ports. These ports allow air and fuel to enter the engine, and exhaust gases to leave after combustion. The cylinder head also holds valves. Intake valves let fresh air or air-fuel mixture come in, and exhaust valves let the burnt gases go out.

Below the cylinder head is the engine block. The engine block is the main body of the engine. Inside it, there are cylindrical holes called cylinders, which are precisely machined. Inside each cylinder, a piston moves up and down.

The piston is connected to the crankshaft through a part called the connecting rod. As you can guess from the name, the connecting rod connects the piston to the crankshaft. The piston moves up and down, and the crankshaft rotates. This connection is very important because vehicles need rotating motion to move the wheels.

Before going further, let’s pause and talk about motion, because this is important. There are two basic types of motion involved here. One is reciprocating motion. This means moving back and forth in the same path, like a syringe or a bicycle pump. The piston moves in this way, up and down inside the cylinder. The second is rotating motion, which is simple rotation, like a ceiling fan or a wheel. The crankshaft rotates.

So the engine’s job is to convert reciprocating motion into rotating motion. That’s one of the key ideas here.

Now here is the simplest way to understand how an engine works. You can think of the engine as a pump. First, it sucks in air from the atmosphere. This air enters through the intake ports in the cylinder head. Fuel is then mixed with this air. This mixing can happen either before the air enters the cylinder or directly inside the cylinder. For this scenario, let’s assume the fuel mixes with air before entering the cylinder.

When the intake valve opens, the air-fuel mixture enters the cylinder. After that, the mixture is compressed, and then it burns. This burning does not happen like an explosion you see in movies. It’s a controlled and rapid burn. This burning creates high pressure, which pushes the piston down. When the piston is pushed down, the connecting rod transfers this motion to the crankshaft, causing it to rotate. That rotating motion is the output power of the engine.

Now this is a good time to explain something called a stroke cycle. You may have heard terms like two-stroke engine and four-stroke engine. For now, we’ll only talk about the four-stroke engine, because that’s what most cars use.

A “stroke” simply means one movement of the piston from top to bottom or bottom to top. In a four-stroke engine, the piston completes four such movements to complete one full cycle. These strokes are intake, compression, power, and exhaust. We won’t go into too much detail just yet, but the idea is that air comes in, gets compressed, burns to produce power, and then the exhaust gases are pushed out. This cycle repeats again and again as long as the engine is running.

Now, earlier we said engines need air and fuel. If you want more power from an engine, there are only two basic ways to get it. One way is to add more fuel. The other way is to add more air. Adding more fuel alone is not efficient, because fuel needs oxygen to burn properly. So the smarter option is to increase the air supply.

So the question becomes, how do we push more air into the engine than it would normally suck in on its own? One way is by using a compressor. A compressor squeezes air and pushes it into the engine at higher pressure. One way to power this compressor is directly from the engine itself using belts. When the engine rotates, it spins the compressor. This setup is called a supercharger, and that itself is a whole topic we can discuss later.

There is another clever way to do this. When an engine runs, a lot of exhaust gases come out of the exhaust pipe. These gases are hot and moving fast, which means they contain energy. Instead of wasting that energy, we can use it to spin a small turbine. If we connect this turbine to a compressor using a shaft, then when the exhaust gases spin the turbine, the compressor also spins. This compressor then sucks in fresh air and pushes it into the engine.

This entire setup is called a turbocharger. The turbine side uses exhaust energy, and the compressor side increases the air going into the engine. This allows the engine to burn more fuel efficiently and produce more power from the same engine size.

And at this point, we've reached a stage where we understand the basic structure of an engine, how motion is created, and how air plays a major role. From here, we can slowly go deeper into stroke-by-stroke operation, differences between petrol and diesel engines, and what actually happens during combustion, but we'll pause here and pick that up next time.

Shock Absorber

Shock absorbers and suspension struts are essential components of your vehicle's suspension system. Their primary purpose is to improve ride comfort while keeping the tires firmly in contact with the road surface for better handling and control. A shock absorber is typically mounted alongside a coil spring or leaf spring, most commonly at the rear of the vehicle. A suspension strut combines both the spring and the shock absorber into a single unit and is usually found at the front. For simplicity, we'll refer to both as dampers.

Whenever your vehicle encounters a bump or pothole, the suspension spring compresses and rebounds to absorb much of the impact before it reaches the vehicle body. Without a damper, however, the spring would continue to bounce. Inside the damper is a piston that moves through oil contained within a sealed cylinder. As the piston travels, oil is forced through specially designed valves that regulate the movement of the suspension.

During minor road irregularities, only a small amount of oil flows through the valves, allowing the suspension to respond smoothly. When larger bumps are encountered, greater oil flow is required, and the valves create increased resistance to control suspension movement and prevent excessive compression or rebound.

Over time, the thin metal valve discs inside the damper gradually fatigue through repeated flexing. As they wear, the damper becomes less effective at controlling suspension movement. Because this deterioration happens gradually, many drivers don't notice the change during everyday driving.

One common symptom of worn dampers is excessive vehicle movement when accelerating or braking. During acceleration, the vehicle may squat at the rear, while braking causes the front to dive more than normal. This extra movement reduces tire contact with the road. In front-wheel-drive vehicles, reduced front tire grip during acceleration may even trigger the traction control system. Increased brake dive can also reduce rear tire stability and contribute to longer stopping distances.

Cornering performance is also affected. Worn dampers allow the body to lean more during turns, making the steering feel less responsive and reducing driver confidence.

As damping performance declines, the wheels continue bouncing after driving over uneven surfaces instead of quickly settling back onto the road. This reduces tire grip, increases tire wear, creates a rougher ride, and can lead to greater driver fatigue during long journeys.

Replacing worn shock absorbers or suspension struts restores proper suspension control, improving ride comfort, vehicle stability, steering response, and overall driving safety.

Brake System

Every moving vehicle contains a large amount of kinetic energy. To bring the vehicle to a stop, that energy must be converted into another form. In conventional petrol and diesel vehicles, the braking system converts this energy into heat through friction. Hybrid and electric vehicles can also recover some of this energy and store it in the battery using regenerative braking.

Most vehicles use disc brakes on each wheel. A disc brake assembly consists of a brake rotor and a caliper fitted with brake pads. When the brakes are applied, the pads clamp onto the spinning rotor, generating friction that slows the vehicle.

Some vehicles still use drum brakes, particularly on the rear wheels. In these systems, hydraulic pressure pushes brake shoes outward against the inside surface of a rotating brake drum to create the friction needed for braking.

Whether the vehicle uses disc brakes, drum brakes, or a combination of both, each brake assembly is connected to the brake master cylinder through hydraulic brake lines filled with brake fluid.

When the driver presses the brake pedal, a push rod moves the pistons inside the master cylinder, creating hydraulic pressure throughout the brake system. This pressure travels through the brake lines and activates the brakes at each wheel.

To reduce pedal effort, most vehicles are equipped with a brake booster positioned between the brake pedal and the master cylinder. The booster uses engine vacuum, along with internal valves and a diaphragm, to multiply the force applied by the driver.

Modern vehicles are also equipped with an Anti-lock Braking System, or ABS. Wheel speed sensors continuously monitor the rotation of each wheel. If the system detects that a wheel is about to lock during hard braking, the ABS controller rapidly adjusts hydraulic pressure to that individual wheel. This repeated release and reapplication of braking force helps maintain steering control while reducing the risk of skidding during emergency stops.

Routine brake inspections include checking the condition and thickness of the brake pads or shoes, measuring rotor and drum wear, inspecting brake hoses and hydraulic lines, and ensuring the brake fluid remains clean and at the proper level. Regular maintenance helps ensure reliable braking performance and safe operation.

Disc Brake

Disc brakes are the most commonly used braking system on modern passenger vehicles. Each wheel is equipped with a brake rotor, a caliper, and a set of brake pads that work together to safely slow and stop the vehicle.

When the driver presses the brake pedal, hydraulic pressure is sent through the brake lines to the caliper. The caliper pistons then force the brake pads against both sides of the spinning brake rotor. The friction generated between the pads and the rotor converts the vehicle's motion into heat, reducing its speed.

Brake pads are manufactured using various friction materials, including ceramic, semi-metallic, and organic compounds. These materials are bonded to steel backing plates that provide structural support. Every time the brakes are applied, a small amount of friction material wears away. Over thousands of braking cycles, the brake rotors also gradually wear as they repeatedly contact the pads.

As brake components wear, braking performance slowly declines. Worn pads may produce squealing noises, while severely worn pads can create grinding sounds as the metal backing plate contacts the rotor. Thin or unevenly worn rotors are more susceptible to thickness variation, which can cause vibration or pulsation when braking.

Other components, such as caliper guide pins, slider bolts, and hydraulic seals, also experience wear over time. If these parts become damaged or seize, they can cause uneven brake pad wear, fluid leaks, reduced braking efficiency, or contamination within the braking system.

Routine brake inspections help identify wear before it becomes a safety concern. Brake pads and rotors should be replaced once they reach the manufacturer's specified minimum thickness. Calipers should also be serviced or replaced whenever worn seals or sticking moving components are found. Keeping the braking system properly maintained ensures safe, consistent, and reliable stopping performance.

Brake Pad

Disc brake systems are fitted to most modern vehicles because they provide reliable and consistent stopping performance. At each wheel, a brake rotor spins with the wheel while a pair of brake pads sit inside the brake caliper. When the brake pedal is pressed, hydraulic pressure forces the caliper to clamp the brake pads against the rotating rotor, creating the friction needed to slow the vehicle.

Brake pads are manufactured from different friction materials, including ceramic, semi-metallic, and organic compounds. Regardless of the material used, every brake pad gradually wears down through normal driving. The rate of wear depends on factors such as driving habits, traffic conditions, vehicle weight, and the type of roads the vehicle is driven on.

As the friction material becomes thinner, braking performance may gradually decline. One of the first warning signs is often a squealing sound during braking. If the pads continue to wear beyond their service limit, the metal backing plate can make direct contact with the rotor, causing grinding noises and damaging the rotor surface. Ignoring worn brake pads can result in more extensive and expensive repairs.

Many modern vehicles are equipped with electronic brake pad wear sensors. When the friction material reaches its minimum service thickness, the system illuminates a warning light on the dashboard to alert the driver that replacement is required. Brake pads are also checked during routine vehicle servicing and should always be replaced before they become excessively worn. On vehicles fitted with electronic wear sensors, the sensor is typically replaced at the same time as the brake pads.

Keeping your brake pads in good condition helps maintain safe braking performance and protects other components of the braking system. If you have any concerns about your vehicle's brakes, consult a qualified service technician.

Drum Brake

Although disc brakes are now used on most passenger vehicles, drum brakes are still commonly found on the rear wheels of many vehicles, particularly older models and some economy vehicles.

A typical drum brake assembly consists of a brake drum, brake shoes, a wheel cylinder, and return springs. The brake drum is mounted to the wheel hub and rotates together with the wheel, while the remaining components are fixed to the backing plate inside the drum.

When the driver presses the brake pedal, hydraulic pressure travels through the brake lines to the wheel cylinder. The pistons inside the wheel cylinder move outward, pushing the brake shoes firmly against the inner surface of the rotating drum. The resulting friction slows the drum and, in turn, the vehicle. Once the brake pedal is released, the return springs pull the brake shoes back into their resting position.

The parking brake also operates through the same brake assembly. Instead of hydraulic pressure, it uses mechanical cables and levers to force the brake shoes against the drum, securely holding the vehicle when parked.

Like all friction materials, the brake shoe lining gradually wears away with normal use. As the lining becomes thinner, you may begin to hear squealing noises during braking. If the friction material wears out completely, the metal brake shoe contacts the drum directly, producing grinding noises while significantly reducing braking performance.

Other drum brake components can also deteriorate over time. Worn or leaking wheel cylinder seals may reduce hydraulic pressure and compromise braking efficiency, while rusted return springs or seized parking brake cables can prevent the brake shoes from operating correctly and lead to uneven or premature wear.

Regular brake inspections help identify these issues before they become serious. Brake shoes should be replaced once they reach the manufacturer's minimum thickness. Leaking wheel cylinders should be repaired or replaced, and any corroded cables or weakened springs should also be serviced to maintain safe and reliable braking performance.

Electronic Parking Brake (EPB)

A parking brake is designed to prevent a vehicle from rolling when it is parked. On most vehicles, it operates on the rear brakes. Traditional parking brakes use either a hand lever or a foot-operated pedal connected by mechanical cables. Many modern vehicles, however, are equipped with an Electronic Parking Brake, or EPB, which is activated using an electrical switch.

When the parking brake switch is pressed, or when the vehicle automatically engages the parking brake, electric motors mounted on the rear brake calipers are activated. These motors rotate threaded mechanisms inside the calipers, driving the pistons forward to clamp the brake pads against the brake rotors and lock the rear wheels. To release the parking brake, the motors simply rotate in the opposite direction, allowing the brake pads to move away from the rotor.

Electronic Parking Brake systems offer several advantages over traditional mechanical designs. Eliminating the handbrake lever creates additional cabin space and removes the need for long mechanical cables running beneath the vehicle. The system also integrates easily with modern driver-assistance features such as Auto Hold, hill-start assist, and automatic parking brake engagement.

Like any electronic system, EPBs can develop faults over time. The most common problems involve the electric motors, electrical wiring, or connector terminals. In many cases, the motor can be replaced separately without replacing the complete brake caliper. Wiring faults are often found where the electrical connector joins the caliper, especially on vehicles regularly exposed to rain, road spray, or harsh environmental conditions.

If the Electronic Parking Brake system detects a fault, it will usually illuminate a warning indicator on the dashboard. Because the system is electronically controlled, accurate diagnosis requires specialized equipment. If a warning light appears or the parking brake does not operate normally, the vehicle should be inspected by a qualified technician to ensure it remains safe and secure when parked.

Brake Master Cylinder

The brake master cylinder is the heart of your vehicle's hydraulic braking system. Its job is to convert the force applied to the brake pedal into hydraulic pressure, allowing the brakes at each wheel to slow and stop the vehicle. It is typically mounted on the firewall in the engine compartment, directly in front of the brake pedal. Attached to the top of the master cylinder is a reservoir that stores the brake fluid used throughout the braking system.

When the driver presses the brake pedal, a push rod transfers that force to the pistons inside the master cylinder. As the pistons move forward, they pressurize the brake fluid. Because brake fluid cannot be compressed, the pressure travels through the brake lines to the brake calipers or wheel cylinders, causing the brakes at each wheel to engage. Once the pedal is released, hydraulic pressure is relieved, allowing the brakes to disengage and the wheels to rotate freely again.

Like any mechanical component, the master cylinder experiences wear over time. The internal rubber seals can gradually deteriorate, allowing brake fluid to bypass the pistons instead of maintaining the pressure needed for effective braking.

Brake fluid can also absorb moisture as it ages. This moisture may cause corrosion inside the master cylinder, damaging the internal surfaces and seals. Regular brake fluid replacement helps minimize corrosion and extends the life of the hydraulic braking system.

A common symptom of internal seal wear is a brake pedal that slowly sinks toward the floor while the vehicle is stopped, such as at a traffic light. In more severe cases, hydraulic pressure can be lost rapidly, causing the pedal to drop suddenly and significantly reducing braking performance.

If you notice changes in brake pedal feel, reduced braking effectiveness, or any warning signs related to the braking system, have your vehicle inspected by a qualified technician as soon as possible.

Blower Motor Resistor

Your vehicle's heating and air conditioning system relies on several components working together to maintain a comfortable cabin temperature. These include the compressor, condenser, expansion device, evaporator, blower motor, refrigerant, and the blower motor resistor or electronic fan control module.

After air passes through the cabin air filter, the blower motor pushes it through the ventilation system and into the passenger compartment. The speed of this airflow is controlled by the blower motor resistor.

When you adjust the fan speed using the climate control panel, the blower motor resistor regulates the amount of electrical current supplied to the blower motor. By varying the current, it allows the fan to operate at different speeds. Many newer vehicles equipped with automatic climate control use a solid-state electronic control module instead of a traditional resistor, providing smoother and more accurate fan speed adjustment.

Over time, the blower motor resistor can fail due to excessive heat, moisture, electrical overload, or the increased current draw caused by a worn blower motor. One of the most common symptoms is the loss of one or more fan speed settings. In many cases, the blower will only operate on its highest speed or may stop working completely.

Other possible faults include damaged electrical connectors, burnt resistor coils, or a failed thermal protection fuse. When these problems occur, airflow through the cabin can be significantly reduced, affecting both heating and air conditioning performance. Defrosting the windshield may also become less effective, reducing visibility during cold or wet weather.

If your climate control system is not operating correctly, a qualified technician can inspect the blower motor, resistor, wiring, and related components to identify the fault and restore proper heating and cooling performance.

Drive-by-Wire (Electronic Throttle Control)

Earlier vehicles used a mechanical cable to connect the accelerator pedal directly to the engine's throttle plate. Pressing the pedal physically opened the throttle, allowing more air into the engine and increasing power output. Most modern vehicles now use an electronic drive-by-wire system, replacing the mechanical cable with sensors, electronic controls, and an electric motor.

When the accelerator pedal is pressed, position sensors measure exactly how far the pedal has moved. This information is sent to the Engine Control Unit, or ECU, which calculates the required throttle opening based on driver input and operating conditions.

The ECU then commands an electric motor inside the throttle body to position the throttle plate accordingly. At the same time, sensors within the throttle body continuously monitor the plate's actual position and send feedback to the ECU, ensuring precise throttle control.

Because the system is electronically controlled, it can work seamlessly with other vehicle technologies such as cruise control, traction control, electronic stability control, and fuel management systems. This results in smoother acceleration, improved fuel efficiency, and better overall vehicle performance.

As with any electronic system, faults can develop over time. Problems may occur within the accelerator pedal sensors, throttle body motor, wiring, or electrical connectors. In addition, carbon deposits can gradually build up around the throttle plate, restricting airflow and preventing smooth throttle movement.

Common symptoms include rough idling, hesitation during acceleration, poor throttle response, reduced engine power, or an illuminated Check Engine light. If the ECU detects a significant fault, it will often place the vehicle into a reduced-power or "limp home" mode. In this condition, the throttle is only allowed to open slightly, enabling the vehicle to be driven carefully to a repair facility while helping to prevent further damage.

If you experience any of these symptoms or notice a warning light on the dashboard, the electronic throttle control system should be inspected using the appropriate diagnostic equipment. Proper diagnosis and repair will restore normal engine performance, drivability, and fuel efficiency.

Spark Plug

Every gasoline or petrol engine relies on controlled combustion to produce power. Inside each engine cylinder, a precise mixture of air and fuel is compressed before being ignited. The component responsible for creating that ignition is the spark plug.

To generate a spark, the ignition coil supplies the spark plug with a high-voltage electrical charge that can reach tens of thousands of volts. This voltage travels through the center electrode to the firing tip.

Older spark plugs typically used copper electrodes, while most modern vehicles are equipped with platinum or iridium-tipped spark plugs. These materials offer greater durability, improved performance, and a longer service life.

The center electrode is surrounded by a ceramic insulator, which prevents the high voltage from escaping through the body of the spark plug. The metal housing is threaded so the spark plug can be securely installed into the cylinder head, positioning the electrodes directly inside the combustion chamber.

At the tip of the spark plug is a small gap between the center electrode and the ground electrode. As the electrical voltage becomes high enough, it jumps across this gap, producing a spark that ignites the compressed air-fuel mixture. This controlled combustion generates the force that drives the engine.

Spark plugs operate in one of the harshest environments found in a vehicle. During combustion, temperatures inside the cylinder can exceed 2,000 degrees Celsius, while the spark plug must also withstand constant pressure and repeated electrical discharge.

With normal use, the electrodes gradually wear, increasing the spark gap and reducing ignition efficiency. Engine problems such as oil entering the combustion chamber or an incorrect air-fuel mixture can also leave carbon deposits on the electrodes. These deposits interfere with spark generation and may result in incomplete combustion or engine misfires.

As ignition performance declines, the engine may consume more fuel, produce less power, idle roughly, or become more difficult to start. In many cases, the engine management system will detect these issues and illuminate the Check Engine light.

Routine maintenance and scheduled spark plug replacement help prevent these problems before they affect engine performance or lead to more costly repairs.

If you notice symptoms such as poor fuel economy, sluggish acceleration, rough engine operation, or an illuminated Check Engine light, have your vehicle inspected by a qualified technician. Replacing worn spark plugs at the recommended service interval helps maintain reliable performance, efficient combustion, and smooth engine operation.

Engine Air Filter

The engine air filter plays an important role in protecting your engine by preventing dust, dirt, pollen, and other airborne contaminants from entering the intake system. Supplying clean air is essential for efficient combustion and long engine life.

Most engine air filters use a pleated filter element. The pleats provide a much larger surface area than a flat filter, allowing a high volume of clean air to flow into the engine while effectively trapping harmful particles before they reach the intake manifold and combustion chamber.

As the filter collects dirt and debris, airflow gradually becomes restricted. Reduced airflow can upset the engine's air-fuel mixture, leading to decreased engine performance, slower acceleration, increased fuel consumption, and rough idling. In some cases, a severely clogged air filter may also contribute to black exhaust smoke due to incomplete combustion.

If a damaged or poorly fitted air filter allows contaminants to bypass the filter element, abrasive particles can enter the engine. Over time, these particles may accelerate wear on internal engine components and reduce engine life.

The engine air filter is inspected during routine vehicle maintenance and should be replaced according to the manufacturer's recommended service schedule or sooner if operating conditions are particularly dusty. Replacing a dirty air filter helps maintain proper airflow, supports better fuel efficiency, and provides reliable protection for your engine.